



Clinical characterization of aortitis and periaortitis: study of 134 patients from a single university hospital

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Abstract

Aortitis and periaortitis refer to the inflammation of the aortic wall and the surrounding tissues. Both conditions are associated with various diseases and express nonspecific manifestations. Early diagnosis and treatment are crucial to improve the prognosis of the disease. This study aimed to assess the causes and main clinical features of aortitis and periaortitis in patients from a single centre in Spain. Observational, retrospective study of patients diagnosed with aortitis or periaortitis at a Spanish referral center over the last decade. 134 patients (87 female; mean age of 55.1 ± 9.1 years) were recruited, 132 of which had aortitis and two periaortitis. Aortitis was associated with giant cell arteritis ($n = 102$), Takayasu's arteritis ($n = 6$), IgG4-related disease ($n = 6$), infectious diseases ($n = 3$), malignancy ($n = 1$), drugs ($n = 1$), isolated aortitis ($n = 1$), and other immune-mediated inflammatory diseases (IMiDs) ($n = 12$). IMiDs included were Sjögren's syndrome ($n = 2$), sarcoidosis ($n = 2$), rheumatoid arthritis ($n = 2$), axial spondyloarthritis ($n = 2$), inflammatory bowel disease ($n = 1$), primary biliary cirrhosis ($n = 1$), idiopathic pulmonary fibrosis ($n = 1$), and polyarteritis nodosa ($n = 1$). Periaortitis was due to idiopathic retroperitoneal fibrosis in both cases. Imaging techniques used for diagnosis included ^{18}F -FDG PET/CT scan ($n = 133$), CT-angiography ($n = 44$), and/or MRI-angiography ($n = 33$). Polymyalgia rheumatica (52.2%) and asthenia (53.7%) were the most common manifestations, followed by limb claudication (23.9%) and inflammatory back pain (26.9%). Acute-phase reactants were typically increased. Aortitis is a common condition and may be associated with multiple non-infectious diseases. Its clinical presentation is often unspecific, requiring a high level of suspicion to get an early diagnosis and treatment.

Keywords Aortitis · Large-vessel vasculitis · Giant cell arteritis · Takayasu arteritis · IgG4-related disease

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Introduction

Aortitis encompasses all the diseases ascribed to inflammation of the aorta, including a wide and heterogeneous group of diseases characterized by inflammatory changes in one or more layers of the aortic wall [1]. Usually, aortitis is of non-infectious inflammatory aetiology, but occasionally can also be of infectious origin. Primary large-vessel vasculitis (LVV), such as giant cell arteritis (GCA), and Takayasu arteritis (TakA) is one of the most common causes of aortitis. There are less frequent aetiologies, which include several types of inflammatory and/or autoimmune diseases [2, 3]. Occasionally, radiotherapy and some drugs can also lead to aortitis, although aortitis can occur isolated too [4–6]. Also, aortitis can be classified into different phenotypes. This can be useful during diagnostic procedures [1]. Systemic inflammatory phenotype, usually associated with

autoimmune conditions, is characterized by systemic manifestations (fever, fatigue, weight loss) and increased acute reactants phase [erythrocyte sedimentation rate (ESR) and/or C-reactive protein (CRP)]. Vascular ischemic phenotype usually indicates vascular occlusion or narrowing, characterized by ischemic features like limb claudication, absent pulses, or stroke. Aneurysmal phenotype, suggesting progression and development of severe complications, is usually presented with localized pain and imaging showing aneurysms or aortic wall thickness. An infectious phenotype has also been described, observed with systemic signs of infection and positive blood cultures. *Treponema pallidum*, *Mycobacterium tuberculosis*, and *Salmonella* spp., among others, are microorganisms with tropism for the aortic vascular endothelium. Finally, there is an imagen-only phenotype characterized by incidental discovery of aortic wall thickening or enhancement on imaging.

Periaortitis is an inflammatory process arising from the adventitia of the aortic wall and extending into the surrounding periaortic space [5]. Several diseases can develop periaortitis, such as idiopathic retroperitoneal fibrosis (RPF) (also called Ormond disease), inflammatory abdominal aortic aneurysm, and histiocytosis, such as Erdheim–Chester disease. Other entities, such as IgG4-related disease (IgG4-RD), can also develop aortitis, periaortitis, and RPF [5, 6]. Key imaging findings, such as the presence of a soft-tissue mantle surrounding the aorta, sparing or minimally involving the wall, can help to diagnose periaortitis.

Indeed, the underlying diseases may be very different and heterogeneous. The clinical presentation is also variable and usually non-specific, making the diagnosis of these conditions truly challenging. Therefore, clinicians must have a high index of suspicion to establish an accurate diagnosis of this entity in a timely way.

In the present report, we aimed to assess the causes and characterize aortitis and periaortitis diagnosed in the last decade in a single referral university hospital.

Patients and methods

Patients, enrollment criteria, and study protocol

We conducted an observational, single-center, retrospective study on 134 patients diagnosed with aortitis or periaortitis. Patients were diagnosed at the Rheumatology Unit from a referral university centre in Northern Spain. The diagnosis of aortitis and periaortitis was usually based on ^{18}F -fluorodeoxyglucose positron emission tomography/computed tomography (^{18}F -FDG PET/CT) scan and sometimes on magnetic resonance imaging angiography (MRI-A) and/or computed tomography angiography (CT-A) scan.

The criteria of the American College of Rheumatology (ACR) of 1990 [7] and/or a positive biopsy of the temporal artery and/or the presence of large-vessel vasculitis when seen in any of the imaging techniques were used to diagnose GCA. Takayasu Arteritis (TakA) was diagnosed based on the ACR 1990 criteria [8] or Ishikawa criteria modified by Sharma et al. [9].

Due to the absence of established criteria for the diagnosis of idiopathic RPF, this disease was diagnosed based on a comprehensive clinical examination using biochemical, immunological, and microbiological tests and imaging methods.

IgG4-RD was diagnosed according to the ACR/EULAR 2020 IgG4-RD Classification Criteria [10], the Umehara criteria [11], the Okazaki criteria [12], and/or expert decision based on clinical, laboratory, and imaging suggestive findings after the exclusion of other causes.

The patients were initially divided into two main groups: (a) patients with aortitis and (b) patients with periaortitis. Subsequently, patients with aortitis were divided according to whether the aortitis was infectious or non-infectious aetiology. Since patients with IgG4-RD can present with both aortitis and/or periaortitis, these were studied in a separate group.

Data collection, clinical definitions, and ethics

Clinical and laboratory data were retrieved according to a pre-established research protocol. Information was stored in a computer database. To minimize entry mistakes, all the databases were double-checked by two blind investigators.

Fever was defined as a temperature of $\geq 38^\circ\text{C}$. Constitutional symptoms were asthenia, anorexia, or weight loss greater than 5% of the normal body weight over the last 6 months before disease diagnosis. Lower limb pain was defined as a diffuse and nonspecific pain involving the thighs. Inflammatory low back pain was diagnosed according to the Assessment of Spondyloarthritis International Society (ASAS) [13].

Arterial hypertension (AHT) was considered when the systolic pressure was > 140 mmHg and/or the diastolic pressure was ≥ 90 mmHg [14]. *Dyslipidemia* was defined if one of the following was present: total cholesterol > 200 mg/dL, triglycerides > 150 mg/dL, high-density lipoprotein (HDL) cholesterol < 40 mg/dL in men or < 50 mg/dL in women, or low-density lipoprotein (LDL) cholesterol > 130 mg/dL.

Routine laboratory markers of disease, including full blood cell count, erythrocyte sedimentation rate (ESR), and serum C-reactive protein (CRP) levels, were obtained at the time of diagnosis and then after each visit of follow-up. ESR was considered to be increased when it was higher than 20 or 25 mm/1st hour for men or women, respectively. A serum

CRP value greater than 0.5 mg/dL was considered abnormal. Anemia was defined as a hemoglobin level < 12 g/dL [15].

The study was approved by the Local Clinical Research Ethics Committee (ethical approval code: 2023.013) and was carried out following the protocol and the standard procedures that ensure compliance with Declaration of Helsinki and Good Clinical Practice standards. According to the Clinical Research Ethics Committee in our center and according to the Spanish local legislation, this type of retrospective research did not require informed consent in our country.

¹⁸F-FDG PET/CT scan acquisition and image analysis

In all patients except one, an ¹⁸F-FDG PET/CT scan was performed in order to confirm the suspicion of aortitis. Before the test, 6 h of fasting were required. Serum glucose level was lower than 160 mg/dL in all patients. 7 MBq/kg of ¹⁸F-FDG were injected and images were taken 180 min after injection, using a Biograph LSO Pico 3D from Siemens Healthcare Molecular Imaging (Hoffman Estates, IL, USA). A CT scan was firstly obtained for attenuation correction and better anatomic localisation, followed by an FDG-PET scan (acquiring 250 s/bed positions). Images were reconstructed using the ordered subsets-expectation maximization (OSEM) algorithm (2 iterations, 8 subsets). Images were visually evaluated by two experienced nuclear medicine specialists according to the intensity of the ¹⁸F-FDG uptake by the vessel wall at the supra-aortic trunks, thoracic aorta, abdominal aorta, iliac arteries, and femoral/tibiofibular arteries.

Statistical analysis

Statistical analyses were conducted using SPSS Statistics for Windows (version 29.0 IBM Corp., Armonk, NY, USA). All continuous variables were tested for normality using the Kolmogorov–Smirnov test. Based on the results of this assessment, variables following a normal distribution were reported as mean $\pm$ standard deviation (SD), whereas non-normally distributed variables were expressed as median and interquartile range (IQR). A p-value < 0.05 was considered as statistically significant.

Results

Overall results

We studied 134 patients (87 female/47 male) with aortitis (n = 132) and/or periaortitis (n = 2). The mean age $\pm$ SD at diagnosis was 55.1 $\pm$ 9.1 years.

The most frequent cardiovascular risk factors were AHT (n = 68; 50.7%), dyslipidemia (n = 64; 47.8%), ischemic

cardiopathy heart disease (n = 26; 17.2%), and diabetes mellitus (n = 17; 12.7%). Thirty-seven (27.4%) patients were smokers or ex-smokers. The main general features of patients included in this study are summarized in Table 1.

Non-infectious aortitis (n = 129) was associated with the following underlying conditions: (a) LVV (n = 108; 83.1%) [GCA (n = 102; 78.5%); TakA (n = 6; 4.6%)]; (b) other IMIDs (n = 12; 9.2%); (c) drugs (n = 1; 0.8%); (d) malignancy (n = 1; 0.8%); (e) isolated idiopathic aortitis (n = 1; 0.8%). Furthermore, we diagnosed 6 (4.6%) patients with aortitis associated with IgG4-RD. Infectious aortitis was observed only in 3 patients (2.3%).

In addition, we observed 2 patients with periaortitis, both in the context of idiopathic RPF.

The diagnosis of aortitis or periaortitis was made by the following imaging techniques: ¹⁸F-FDG PET/CT scan (n = 133; 99.3%), CT-A scan (n = 44; 32.8%), and MRI-A (n = 33; 24.6%). Patients with suspicion of aortitis demonstrated by CT-A scan or MRI-A underwent an additional ¹⁸F-FDG PET/CT to confirm the diagnosis.

Regarding the anatomic location of aortitis, the different aortic segments affected were: ascending thoracic aorta (n = 105; 78.4%), descending thoracic aorta (n = 79; 58.9%), aortic arch (n = 67; 50%), and abdominal aorta (n = 49; 36.6%).

Six (4.5%) patients developed aortic aneurysms. Five patients developed an aneurysm during the evolution of aortitis (one patient 1 month after diagnosis, one patient 1 year, two patients 3 years, and one patient 7 years later). There was only one patient who was diagnosed with aortic aneurysm before diagnosis of aortitis. The location of aneurysms was abdominal aorta (n = 4; 66.7%) and thoracic aorta (n = 2; 33.3%). The underlying diseases were GCA (n = 5; 83.3%) and TakA (n = 1; 16.6%).

Subtypes of aortitis

Non-infectious aortitis

Large-vessel vasculitis (i) Giant cell arteritis We diagnosed 102 patients (65 women and 37 men) with aortitis related to GCA. The mean age $\pm$ SD of these patients was 67.9 $\pm$ 9.9 years. The median [IQR] time from GCA diagnosis to aortitis diagnosis was 3 [2–6] months. The most frequent systemic manifestations at aortitis diagnosis were polymyalgia rheumatica (PmR) and constitutional syndrome observed in 63 (62.8%) and 53 (52%) patients, respectively. Among ischemic manifestations, headache was the most common, objectified in 40 (39.2%) patients. Inflammatory back pain and limb claudication were observed in 29 (28.1%) and 26 (25.5%) patients, respectively. The median [IQR 25th–75th] of serum CRP and ESR values at diagnosis were 0.6 [0.4–3.2] mg/dL and 26.5 [7–54] mm/1st

Table 1 General features of 134 patients with aortitis and periaortitis

	AORTITIS						PERIAORTITIS			
	Overall n = 134	GCA n = 102	TakA n = 6	Immune-mediated diseases n = 12	IgG4-RD n = 6	Malignancy n = 1	Drugs n = 1	Isolated n = 1	Infectious n = 3	Retropertitoneal idiopathic fibrosis n = 2
Sex (female/male)	87F/47 M	65F/37 M	6F	10F/2 M	3F/3 M	1 M	1F	1 M	3 M	2F
Age (mean $\pm$ SD)	55.06 $\pm$ 9.08	67.9 $\pm$ 9.9	41.8 $\pm$ 14.1	61.3 $\pm$ 16	55.3 $\pm$ 11.1	62	48	46	49.7 $\pm$ 17	63.5 $\pm$ 3.5
Cardiovascular risk factors, n (%)										
AHT	68 (50.7)	54 (52.9)	4 (66.6)	7 (58.3)	3 (50)	–	–	–	–	–
DLP	64 (47.8)	48 (47.1)	4 (66.6)	6 (50)	2 (33.3)	–	1 (100)	1 (100)	1 (33.3)	1 (50)
DM	17 (12.7)	14 (13.7)	–	1 (8.3)	–	1 (100)	1 (100)	–	–	–
Smokers/ex-smokers	37 (27.4)	23 (22.6)	5 (83.3)	6 (50)	1 (16.7)	–	–	1 (100)	–	1 (50)
Diagnosis of aortitis (imaging techniques)										
CT-A scan	44	25	5	5	3	1	1	1	1	2
18 F-FDG PET/CT scan	133	102	6	12	6	1	1	1	2	2
MRI-A	33	20	4	6	1	–	–	–	–	2
Delayed diagnosis of aortitis (months between clinical onset and diagnosis of aortitis) median [IQR]	9 [7.5–10]	3 [2–6]	7.5 [2–20.25]	3 [2–10]	9 [3.75–21]	32	10	8	13 [7.5–18.5]	9.5 [8.25–10.75]
Systemic manifestations, n (%)										
General syndrome	72 (53.7)	53 (52.0)	3 (50)	7 (58.3)	4 (66.7)	1 (100)	1 (100)	1 (100)	1 (33.3)	1 (50)
Fever	22 (16.4)	18 (17.7)	1 (16.7)	3 (25)	–	–	–	–	–	–
PnR	70 (52.2)	63 (62.8)	–	3 (25)	3 (50)	1 (100)	–	–	–	–
Cranial ischemic manifestations, n (%)										
Headache	43 (32.1)	40 (39.2)	2 (33.3)	–	1 (16.7)	–	–	–	–	–
Jaw claudication	12 (8.9)	12 (11.8)	–	–	–	–	–	–	–	–
Ocular manifestations	20 (14.9)	17 (16.7)	1 (16.7)	1 (8.3)	1 (16.7)	–	–	–	–	–
Temporal artery abnormalities	11 (8.2)	11 (10.8)	–	–	–	–	–	–	–	–
Other manifestations, n (%)										
Limb claudication	32 (23.9)	26 (25.5)	3 (50)	1 (8.3)	1 (16.7)	1 (100)	–	–	–	–
Abdominal pain	8 (6.0)	3 (2.9)	1 (16.7)	2 (16.6)	1 (16.7)	–	–	–	–	1 (50)
Inflammatory back pain	36 (26.9)	29 (28.1)	2 (33.3)	2 (16.6)	3 (50)	–	–	–	–	–
Laboratory										
CRP (mg/dL), median [IQR]	0.4 [0.4–4.0]	0.6 [0.4–3.2]	0.4 [0.2–2.1]	0.4 [0.9–3.6]	3.3 [0.9–4.2]	0.4	1.1	0.5	0.4	0.4
ESR (mmHg/1st h), median [IQR]	27.8 [15.5–39.5]	26.5 [7–54]	16.5 [3.5–32.5]	36 [6–47]	53 [9–65]	–	50	29	7 [3–11]	12.5 [10–15]
Hb (g/dL), mean $\pm$ SD	13.3 $\pm$ 1.2	12.8 $\pm$ 1.5	12.8 $\pm$ 0.9	13.2 $\pm$ 1.6	14.1 $\pm$ 1.5	14.9	11.1	14.7	14.3 $\pm$ 4.1	13.1 $\pm$ 1.25

AHT arterial hypertension, CRP C-reactive protein, CT-A computed tomography angiography, DLP dyslipidaemia, DM diabetes mellitus, ESR erythrocyte sedimentation rate, GCA giant cell arteritis, Hb haemoglobin, IgG4-RD IgG4-related disease, MRI-A magnetic resonance imaging angiography, 18 F-FDG PET/CT 18 F-fluorodeoxyglucose positron emission tomography/computed tomography, PnR polymyalgia rheumatica, TakA Takayasu's arteritis

hour, respectively. The ascending thoracic aorta was the most commonly affected aortic segment, followed by the descending thoracic aorta (Fig. 1).

Five patients developed aneurysms as a complication of aortitis, located in the abdominal aorta in three patients and descending thoracic aorta in two patients.

(ii) *Takayasu Arteritis* We diagnosed 6 patients (all women) with aortitis associated with TakA. The mean age of these patients was 41.8 ± 14.14 years.

The most frequent clinical manifestations were asthenia ($n=4$; 66.6%) and limb claudication ($n=3$; 50%). The median [IQR 25th–75th] of serum CRP and ESR values at diagnosis were 0.4 [0.2–2.1] mg/dL and 16.5 [3.5–32.5] mm/1st hour, respectively. The aortic segments most frequently affected were the ascending thoracic aorta ($n=4$; 66.6%) and aortic arch ($n=4$; 66.6%) (Fig. 1). A 42-year-old woman developed abdominal aneurysm as a complication of aortitis. The main clinical and laboratory features of these patients are summarized in Table 2.

Aortitis related to other immune-mediated inflammatory diseases We identified 12 patients (10 women/2 men) with aortitis related to different IMIDs. The mean age was 61.3 ± 16 years. The underlying IMIDs were Sjögren syndrome ($n=2$), sarcoidosis ($n=2$), axial spondylarthritis ($n=2$), rheumatoid arthritis ($n=2$), idiopathic pulmonary

fibrosis ($n=1$), polyarteritis nodosa ($n=1$), biliary primary cirrhosis ($n=1$) and ulcerative colitis ($n=1$).

The main clinical and laboratory features of these patients are summarized in Table 3.

Malignancy A 62-year-old male with a gastric gastrointestinal stromal tumor (GIST) was diagnosed with aortitis during an oncology follow-up 27 months after the malignancy diagnosis. A CT was performed, and a concentric increase of the aortic arch wall was observed. An ^{18}F -FDG PET/CT was also performed, revealing ^{18}F -FDG uptake in the aortic arch wall. The patient was referred to constitutional syndrome, with a CRP value of 0.5 mg/dL. The main clinical manifestation was constitutional syndrome related to the neoplasia, and the patient did not receive treatment for aortitis because there were no signs of active vasculitis.

Drugs We also present a case of subclinical aortitis related to cancer immunotherapy, which we have previously reported [16]. Briefly, this is a case of a 48-year-old female who developed aortitis after starting intravenous nivolumab (3 mg/kg every other week), an IgG4 programmed cell death 1 (PD-1) inhibitor, due to a malignant melanoma with metastatic lesions. The ^{18}F -FDG PET/CT scan showed FDG uptake in the thoracic aortic wall. The patient had no clinical signs of GCA, and acute phase reactants were normal. As the patient was asymptomatic for aortic involvement and

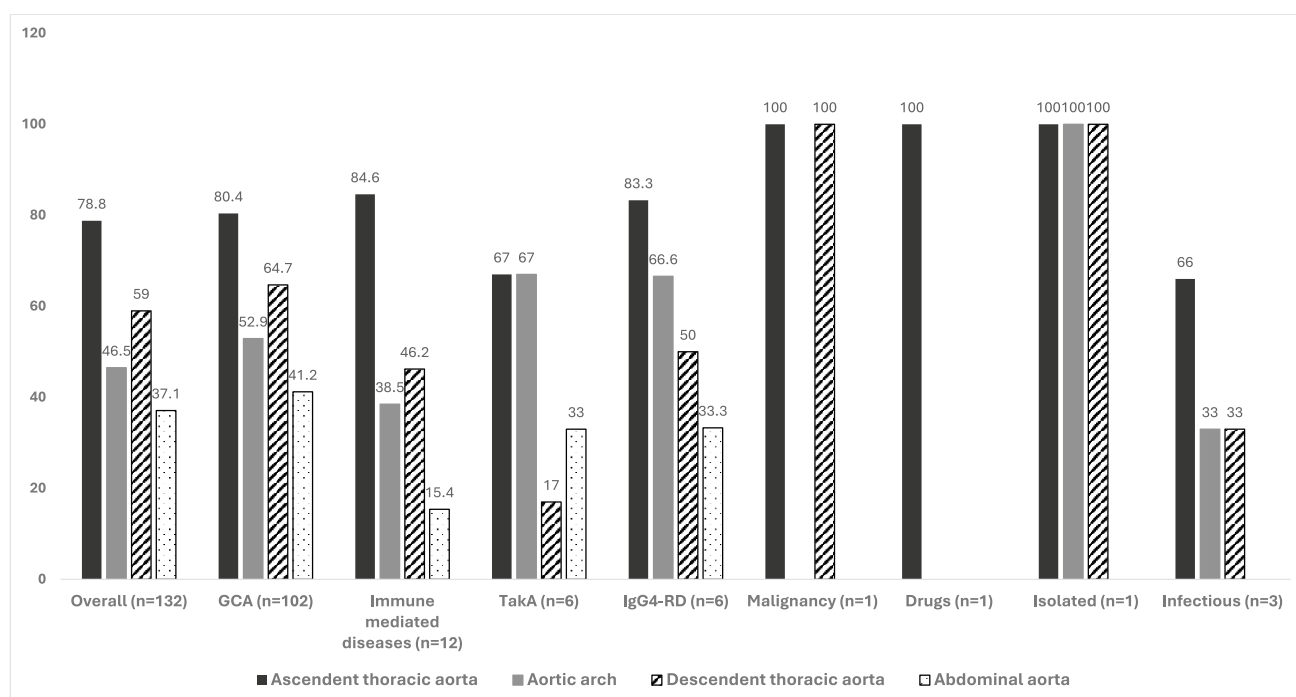


Fig. 1 Aortic segments involved in the 132 patients with aortitis. GCA giant cell arteritis, IgG4-RD IgG4-related disease, TakA Takayasu's arteritis

Table 2 General features of 6 patients with aortitis related to TakA

Case	Age/sex	Clinical manifestations	Affected segments of the aorta	CRP (mg/dL)/ ESR (mmHg 1st hour)	Hemo- globin (g/ dL)
1	52/F	Asthenia, anorexia, upper limb claudication, inflammatory back pain, blood pressure difference between upper limbs	Ascending thoracic aorta, aortic arch	0.5/59	13.7
2	50/F	Lower limb claudication	Ascending thoracic aorta, aortic arch, abdominal aorta	<0.1/5	13.5
3	42/F	Asthenia, weight loss, limb claudication, inflammatory back pain	Ascending thoracic aorta, aortic arch	<0.4/2	11.6
4	44/F	Asthenia, weight loss, fever, abdominal pain, visual symptoms, headache	Ascending thoracic aorta, aortic arch, descending thoracic aorta	0.1/3	13.4
5	49/F	Headache	Abdominal aorta, supraaortic trunks	2.6/28	13.3
6	14/F	Asthenia	Supraaortic trunks, subclavian artery	6/34	11.5

CRP C-reactive protein, ESR erythrocyte sedimentation rate, TakA Takayasu's arteritis

acute phase reactants were normal, no specific treatment for aortitis was initiated.

Isolated aortitis We present a 46-year-old male with polyarthralgia and abdominal pain with acute phase reactants elevated (CRP of 0.5 mg/dL and ESR of 59 mm/1st hour). A complete biochemistry, blood count, immunological, and microbiological study was carried out, ruling out underlying conditions without fulfilling GCA or TakA criteria. The ^{18}F -FDG PET/CT scan showed increased metabolic activity in the thoracic aorta and supra-aortic trunks. The patient was diagnosed with isolated aortitis. Nowadays, the patient is under treatment with corticosteroids with the absence of inflammatory activity inside the aorta through an ^{18}F -FDG PET/CT control.

IgG4-related disease Six patients (3 women/3 men) of our series were diagnosed with aortitis related to IgG4-RD (Table 4). The mean age was 55.3 ± 11.1 years. Aortitis was diagnosed at the same time as IgG4-RD diagnosis in five of them; the other one developed aortitis 5 years after the diagnosis of IgG4-RD. The most frequent clinical manifestations were general syndrome ($n=4$; 66.7%), PmR ($n=3$; 50%), and inflammatory back pain ($n=3$; 50%). The median [IQR] of serum CRP and ESR values were 3.3 [0.9–4.2] mg/dL and 53 [9–65] mm/1st hour, respectively. Besides the aorta and its major branches, other organs were affected in these patients: esophagus ($n=1$; 16.6%), larynx ($n=1$; 16.6%), mediastinum ($n=1$; 16.6%), lungs ($n=1$; 16.6%) and nervous central system ($n=1$; 16.6%).

Infectious aortitis

We observed three patients with infectious aortitis due to syphilis, Lyme disease, and fungal aortitis.

A 66-year-old male was diagnosed with syphilitic aortitis by ^{18}F -FDG PET/CT scan performed during hospital admission for general syndrome and PmR. Syphilis had been diagnosed 5 years before due to skin lesions with a positive biopsy for syphilitic roseola. Serum CRP and ESR values were 0.4 mg/dL and 11 mm/1st hour, respectively. At the diagnosis of syphilis, the rapid plasma regain (RPR) was positive at titre 1/16 and *Treponema pallidum* hemagglutination assay (TPHA) was also positive at titre 1/2560. The patient was treated with benzathine penicillin.

When aortitis appeared, the patient received oral glucocorticoids and methotrexate. Today, the patient is asymptomatic from a vascular viewpoint and without treatment.

We also present a 51-year-old male diagnosed with Lyme disease with neurological involvement (subjective memory complaints). Eleven months after the *Borrelia burgdorferi* infection, the patient developed a general syndrome, arthralgias, and acute kidney failure. An ^{18}F -FDG PET/CT scan revealed FDG uptake and increased metabolic activity in the aortic thoracic and abdominal aorta, as well as supra-aortic trunks. The patient received antibiotic therapy with intravenous ceftriaxone at a dose of 2 g/day for 28 days with partial improvement.

Finally, we present the case of a 32-year-old male with acute myeloblastic leukaemia who was admitted to the hospital with a decreased level of consciousness. A cranial CT scan showed brain lesions suggestive of septic abscesses. The patient was treated vigorously with intravenous antibiotic therapy but died. Necropsy revealed fungal involvement of the aorta by *Scedosporium prolificans* with multiple septic emboli.

Table 3 General features of 12 patients with aortitis related to immune-mediated inflammatory diseases

Case	Associated disease	Age/Sex	CVRF	Clinical manifestations	Affected segments of aorta	CRP (mg/dL)/ESR (mmHg 1st hour)	Haemoglobin (g/dL)
1	Axial spondylarthritis	74/F	AHT, DM, DLP	Fever, general syndrome, PmR, upper and lower limb claudication	Ascending thoracic aorta	3.3/46	14.7
2	Axial spondylarthritis	84/F	AHT, DLP	General syndrome, inflammatory back pain, shoulder pain	Supraaortic trunks, femoral artery	0.1/3	11.4
3	Idiopathic lung fibrosis	55/F	AHT, DLP	Abdominal pain, inflammatory back pain	Ascending thoracic aorta, aortic arch, supraaortic trunks, descending thoracic aorta, abdominal aorta	0.16/2	14.1
4	Primary biliary cirrhosis	50/F	–	Fever, knee pain, ankle pain	Ascending thoracic aorta, aortic arch, descending thoracic aorta	1.2/9	13.8
5	Rheumatoid arthritis	62/F	–	General syndrome, voice alteration	Ascending thoracic aorta, aortic arch	2.2/40	–
6	Rheumatoid arthritis	62/F	–	PmR, lower limb pain	Ascending thoracic aorta, aortic arch, supraaortic trunks	1.5/50	–
7	Sarcoidosis	37/M	DLP	Fever, general syndrome, dyspnoea	Femoral artery	0.9/48	13.3
8	Sarcoidosis	58/F	AHT	General syndrome, PmR, lower limb pain	Ascending thoracic aorta, aortic arch, supraaortic trunks	<0.4/2	15.4
9	Sjögren syndrome	73/F	AHT, DLP	General syndrome, visual alterations	Ascending thoracic aorta, supraaortic trunks	0.9/111	11.6
10	Sjögren syndrome	79/F	AHT, DLP	General syndrome, dry syndrome	Ascending thoracic aorta	4/92	10.6
11	Testicular PAN	32/M	–	Fever, testicle pain, skin injuries	Ascending thoracic aorta, aortic arch, supraaortic trunks, descending thoracic aorta	0.4/12	14.7
12	Ulcerative colitis	69/F	AHT	Abdominal pain	Descending thoracic aorta, abdominal aorta	<0.4/36	14

CRP C-reactive protein, CVRF cardiovascular risk factors, DLP dyslipidaemia, DM diabetes mellitus, ESR erythrocyte sedimentation rate, F female, M male, PmR polymyalgia rheumatica

Periaortitis

Two of our 134 patients (1.5%) had periaortitis related to idiopathic RPF. One of the patients was a 61-year-old female with limb claudication and abdominal pain who underwent a CT scan during a routine urological check-up for renal lithiasis. This CT showed retroperitoneal tissue fibrosis. Then, an ^{18}F -FDG PET/CT was performed, demonstrating increased activity in the aortic wall.

The other patient was a 66-year-old female who was diagnosed with idiopathic RPF during a study of the constitutional syndrome and acute kidney failure. Ultrasound

demonstrated ureteral obstruction by an extrinsic compression. Subsequent studies with CT and ^{18}F -FDG PET/CT showed RPF with increased metabolic activity in the infra-renal aorta.

Discussion

In this article, we performed a clinical characterization and imaging diagnosis of aortitis and periaortitis from a reference single University Hospital in Spain.

Table 4 General features of 6 patients with aortitis related to IgG4-RD

Case	Age/Sex	CVRF	Diagnosis of IgG4-RD	Diagnosis of aortitis	Clinical manifestations	CRP (mm/dL)/ESR (mmHg 1st hour)	Level of IgG4 (mg/dL)/plasma blast (cells/mL)	Affected organs
1	57/F	–	Laboratory variables + PET + clinical manifestations ACR criteria	MRI, ¹⁸ F-FDG PET/CT scan	General syndrome, PmR, inflammatory back pain	0.4/9	125/1218	Esophagus Larynx: subglottic stenosis Aorta (thoracic aorta and aortic arch)
2	46/M	AHT	PET ACR criteria	¹⁸ F-FDG PET/CT scan	Constitutional syndrome, headache, inflammatory back pain	2.6/6	105.8/87	Aorta (thoracic aorta and aortic arch)
3	42/M	DLP	PET	CT scan, PET/CT	Constitutional syndrome	5.5/65	–	Aorta (thoracic aorta and aortic arch)
4	51/F	–	Laboratory variables + PET ACR criteria	¹⁸ F-FDG PET/CT scan	Constitutional syndrome, PmR, limb claudication	0.1	4.8/787	Aorta (thoracic aorta)
5	70/M	AHT	Biopsy + PET Okazaki criteria	CT scan, ¹⁸ F-FDG PET/CT scan	Inflammatory back pain, abdominal pain	4/53	40.3/–	Pachymeningitis, Mediastinum adenopathies, Lungs: infiltration Aorta (abdominal aorta)
6	66/F	AHT, DLP	Laboratory variables + PET	CT scan, MRI, ¹⁸ F-FDG PET/CT scan	PmR	4.3/95	51.4/3299	Aorta (thoracic and abdominal aorta, and aortic arch)

ACR American College of Rheumatology, AHT arterial hypertension, CRP C-reactive protein, CT computed tomography, CVRF cardiovascular risk factors, DLP dyslipidaemia, DM diabetes mellitus, ESR erythrocyte sedimentation rate, ¹⁸F-FDG PET/CT ¹⁸F-fluorodeoxyglucose positron emission tomography/computed tomography, IgG4-RD IgG4-related disease, MRI magnetic resonance image, PET positron emission tomography, PmR polymyalgia rheumatica

Aortitis presents a wide and heterogeneous range of symptoms, often non-specific, which, in many cases, unfortunately, delays the diagnosis of the disease. Hence, a high index of suspicion for aortitis is always required. Apart from fever, headache, inflammatory back pain, and PmR symptoms, aortic insufficiency, aortic aneurysms, and aortic dissection have been described [17, 18].

Understanding aortitis pathophysiological mechanisms could be difficult because of the multiple underlying causes of aortitis above mentioned. According to this, inflammation of the aorta can be originated by different mechanisms. Regarding GCA, T-cell-mediated immune response targets the tunica media, leading to granulomatous inflammation with giant cells, and destruction of elastic fibers results in vessel wall weakening, leading to aneurysms or rupture of aorta and its mayor branches. In TakA, there is also a T-cell-mediated immune response and granuloma formation, chronic inflammation leading to narrowing and stenosis or occlusion of great vessels. On the other hand, in aortitis

related to IgG4 disease, infiltration of IgG4-positive cells causes thickening of the aorta wall. Regarding infectious aortitis, bacterial invasion of the aortic wall leads to neutrophilic infiltration, necrosis, and vascular weakening.

Imaging techniques, such as ¹⁸F-FDG PET/CT scan, MRI-A, CT-A, and ultrasound (with limited access to the thoracic aorta), have been the most critical diagnostic and monitoring tool in LVV [19, 20]. However, PET images are generally quite similar regardless of the underlying pathology. Aortitis can affect any segment of the aorta, but certain locations are more commonly involved depending on the underlying cause. Takayasu's arteritis often affects the aortic arch and its main branches, as well as the thoracic and abdominal aorta. On the other hand, giant cell arteritis primarily involves the ascending aorta and the arch. Syphilitic aortitis typically affects the ascending aorta and aortic root [1]. In our study, the ascendent and descendent thoracic aorta are the most frequently involved parts of the aorta, except in Takayasu arteritis and IgG4-RD, where the

ascending thoracic aorta is the most frequently involved followed by aortic arch.

Therefore, the diagnosis of aortitis relies on a combination of clinical expertise and interpretation of imaging and laboratory findings (Fig. 2). Due to an often non-specific clinical presentation, we proposed a flowchart for aortitis diagnosis: with a high index of suspicion (polymyalgic syndrome, constitutional syndrome, unknown origin fever) and elevated acute phase reactants, an imaging technique should be performed. Once aortitis is demonstrated by image techniques, medical history, physical examination, and laboratory findings (immunological markers, microbiological studies), help us to investigate if aortitis is idiopathic or associated with other underlying diseases.

In our series, the most frequent clinical features included constitutional syndrome (53.7%), PmR (52.2%), headache (32.1%), and limb claudication (23.9%). Among non-infectious aortitis, the two most frequent causes were the LVV-GCA and TakA. While both diseases may show significant overlap in the pathological findings, their epidemiological patterns and clinical presentations are quite different. Both are associated with an inflammatory cellular infiltrate in the aortic media, adventitia, and vasa vasorum that contains a predominance of lymphocytes, macrophages, and multinucleated giant cells, as well as

the destruction of the elastic lamina [1, 4, 21]. TakA often shows contiguous involvement, while GCA is characterized by focal arterial inflammation with skip lesions. The mural thickness is usually greater in TakA [1, 22, 23]. GCA is more common in Caucasian adults older than 50 years, while TakA is more often presented in young women, mainly from Asia, Africa, and Latin America [24, 25].

Although GCA was traditionally considered to be a disease of the branches of the carotid arteries, the advance and increased use of imaging techniques has led to a better understanding of this disease, with up to 45–60% of GCA patients showing involvement of the aorta and/or its main branches [18, 26, 27]. GCA associated with aortitis affects younger people than classical cranial GCA [27]. Aortic involvement is often heralded by constitutional symptoms. Aortitis in GCA patients may lead to aortic aneurysms with a risk of rupture or aortic dissection. Evans et al. observed that GCA patients had a 17-fold increased risk of thoracic aortic aneurysm and a 2.4-fold increased risk of abdominal aortic aneurysms [28].

Most cases of TakA involve inflammation of one or more aortic segments, often in combination with branches, mainly subclavian, carotid, or renal arteries. The development of arterial stenosis and aneurysms leads to claudication, bruits,

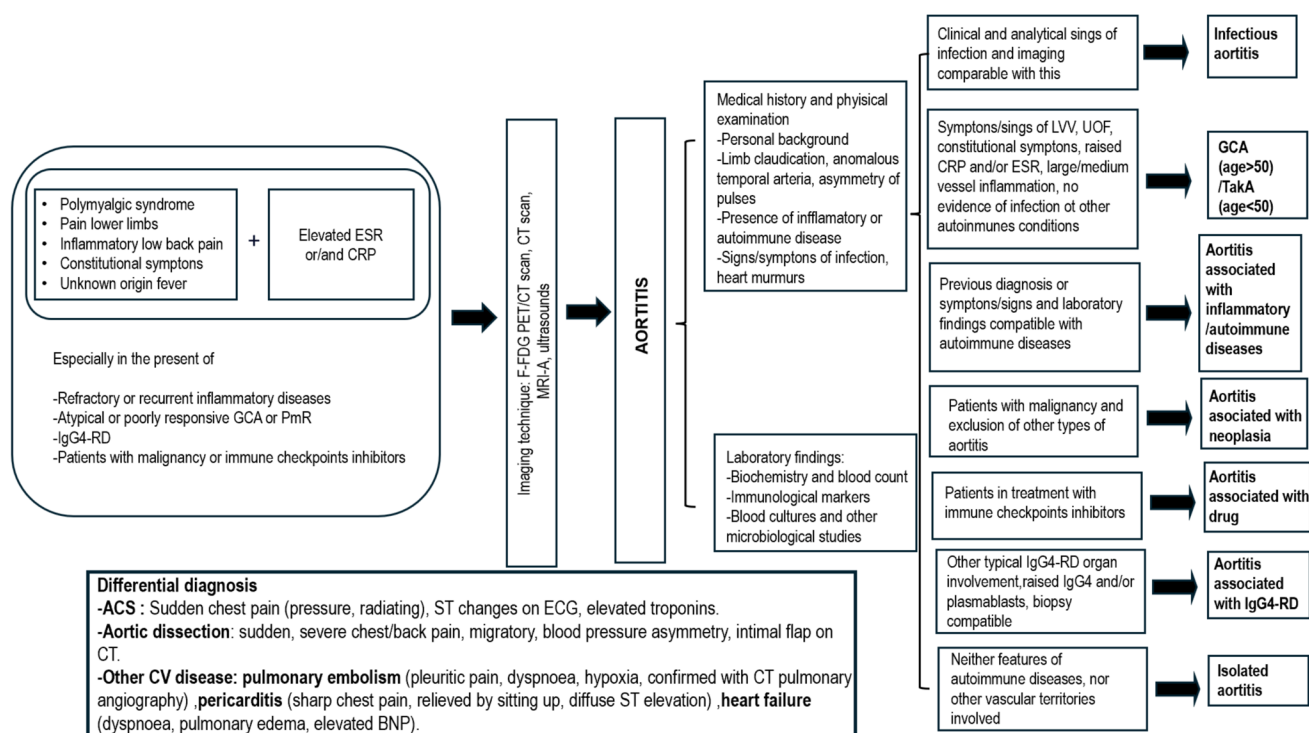


Fig. 2 Proposed flow-chart for aortitis diagnosis. *CRP* C-reactive protein, *CT* computed tomography, *ESR* erythrocyte sedimentation rate, ^{18}F -*FDG PET/CT* ^{18}F -fluorodeoxyglucose positron emission tomography/computed tomography, *GCA* giant cell arteritis, *IgG4-RD* IgG4

related disease, *LVV* large vessel vasculitis, *MRI* magnetic resonance image, *PET* positron emission tomography, *PmR* polymyalgia rheumatica, *UOF* unknown origin fever

limb pain, and diminished or even absent pulses [25]. Constitutional symptoms are also frequent.

Aortitis may also be associated with other diseases and should be suspected in patients with various multi-vessel vasculitis, in autoimmune or inflammatory diseases, and presenting with any clinical feature that may suggest aortic involvement.

Aortitis, like other vasculitides, can also occur as a paraneoplastic manifestation [29]. Aortitis can also be secondary to drugs, especially immune checkpoint inhibitors, which are increasingly used in Oncology [16, 30, 31].

Sometimes, aortitis may occur without clinical or histopathological evidence of systemic disease or involvement of other vascular territories, which is known as isolated aortitis. It is more frequent in women with a mean age between 63 and 72 years [5, 24, 32]. It is usually asymptomatic, although it may lead to thoracic aortic aneurysms, aortic dissection, or aortic valve regurgitation. This entity is often an incidental finding or is diagnosed when complications arise.

Infectious aortitis describes an infection of the aorta wall. Aortic infections arise from either contiguous spread from adjacent structures or septic emboli and hematogenous spread of microorganisms to the aortic wall through a vulnerable plaque or preexisting aneurysm [1, 33]. Although the aorta is normally very resistant to infection, the existence of an abnormal aortic wall, such as in atherosclerotic disease, preexisting aneurysm, post-medical device placement, or post-surgery, makes it more susceptible to infection [34]. *Staphylococcus aureus*, *Pneumococcus*, *Escherichia coli*, and *Salmonella* are the pathogens identified in most reports [33]. Syphilitic aortitis typically appears several years after systemic *Treponema pallidum* infection. It is associated with a chronic inflammatory infiltrate of the medial and adventitial vasa vasorum, leading to medial necrosis and wrinkled appearance of the intima histologically known as “tree bark-ing” [35]. Although syphilis was a common cause of aortitis a few decades ago, nowadays, it is a rare entity, and there was only one case in our series. Fungal aortitis from *Candida* or *Aspergillus* and tuberculous aortitis are uncommon and typically arise in immunocompromised hosts [1], such as the patient of our series with acute myeloblastic leukaemia. Lyme disease can affect almost any organ, including the aorta. In our series, one patient had an aortitis due to *Borrelia burgdorferi* infection.

Periaortitis is a rare condition that involves both the aortic wall and the surrounding soft tissues. Several entities, such as idiopathic RPF, inflammatory abdominal aortic aneurysm, as well as Erdheim-Chester disease, may occur with periaortitis [5]. In our series, we only observed two patients with periaortitis, both in the context of an idiopathic RPF. The incidence of idiopathic RPF is 0.1–1.3/100,000 person-years and often involves the adventitia of the abdominal aorta, the iliac arteries, and the surrounding retroperitoneum. Ureteral

involvement is the most common disease-related complication, which can lead to obstruction of the pelvic ureteral tract [5, 36].

Inflammatory abdominal aortic aneurysms account for 3–10% of all abdominal aortic aneurysms. This entity is associated more strongly with tobacco smoking, younger age at presentation, and family history of aortic aneurysm [37, 38].

Erdheim–Chester disease is a rare histiocytic disorder with diverse clinical manifestations, ranging from indolent, localized presentations to life-threatening, multisystem disease. A series of recurrent activating kinase mutations and fusions involving the canonical MAPK (RAS-RAF-MEK-ERK) and phosphatidylinositol 3-kinase (PI3K)-AKT pathways have been discovered in a large proportion of patients with this disease [39]. Histological findings often include fibro-inflammatory infiltrates containing foaming-activated histiocytes, usually accompanied by Touton giant cells [40]. On immunohistochemical staining, the histiocytes of this disease are positive for CD68, CD163, and factor XIIIa and negative for CD1a and CD207. Positivity for S100 is rarely observed [41]. The periaortic fibrosis that can be seen in Erdheim–Chester disease may affect the entire length of the aorta, wrapping around it and giving rise to what is known as “coated aorta” on CT scans or MRI [42].

IgG4-RD is a systemic immune-mediated fibroinflammatory disease that can affect almost any organ [43, 44]. The diagnosis of the disease may be challenging and should be based on specific histopathological findings, typical laboratory and radiological aspects, and an appropriate clinical context. IgG4-RD can present with aortitis, periaortitis, and retroperitoneal fibrosis [45]. The aortitis in the setting of an IgG4-RD affects about 8% of patients with this disease, representing 7–9% of the non-infectious thoracic aortitis [45]. The thoracic aorta involvement is twice as common as abdominal involvement. The presence of inflammatory thoracic aortic aneurysm is frequent in these patients [45]. Periaortitis is present in 20–36% of the patients with IgG4-RD, the abdominal aorta, the segment more usually involved, mainly the infra-renal region, along with the iliac arteries [45]. In our series, we observed six patients with aortitis related to IgG4-RD, but no aneurysms were observed in any of them.

In summary, aortitis represents an inflammation of the aortic wall that can be idiopathic or associated with a wide range of infectious and non-infectious diseases, being non-infectious, mainly GCA, the most common cause in our setting. Clinical presentation is often non-specific, and it is important to have a high index of suspicion, especially when general syndrome, fever, and systemic manifestations are present and other diagnoses such as infection or malignancy have been excluded. Imaging techniques play a central role in identifying the etiology, extent, and activity of disease.

Our study has several limitations derived from its retrospective nature. The main limitations include the potential for selection bias, as it relies on pre-existing data that may not represent the broader population and the possibility of missing or incomplete data due to the uneven quality of historical records. Additionally, the inability to control variables prospectively can introduce confounding factors, making it difficult to establish causal relationships and limiting the search for associations. Moreover, limited geographic and demographic diversity in aortitis research can lead to the underrepresentation of certain populations, preventing the understanding of how the disease affects different ethnic groups and regions.

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Data availability statement Data is available upon reasonable request by any qualified researchers who engage in rigorous, independent scientific research and will be provided following review and approval of a research proposal and Statistical Analysis Plan (SAP) and execution of a Data Sharing Agreement (DSA). All data relevant to the study are included in the article.

Declarations

Conflict of interest Disclosures that might be interpreted as constituting possible conflict(s) of interest for the study. Javier Loricera had consultation fees/participation in company-sponsored speaker's bureau from Roche, Galápagos, Novartis, UCB Pharma, MSD, Celgene, Astra Zeneca, and Grünenthal. He received support for attending meetings and/or travel from Janssen, Abbvie, Roche, Novartis, MSD, UCB Pharma, Celgene, Lilly, Pfizer, and Galápagos. Santos Castañeda has received research support from MSD and Pfizer and had consultation fees/participation in the company-sponsored speaker's bureau from BMS, Eli-Lilly, MSD, Roche, and UCB. S. Castañeda is also assistant professor of the cátedra EPID-Future, funded by UAM-Roche, Universidad Autónoma de Madrid (UAM), Spain. Ricardo Blanco received grants/research support from AbbVie, MSD, and Roche. He also received consultation fees/participation in a company-sponsored speaker's bureau from AbbVie, Pfizer, Roche, Bristol-Myers, Janssen, Lilly, UCB, and MSD. The following authors did not declare financial disclosure: Carmen Secada-Gómez, Adrián Martín, Fernando López Gutiérrez, Lucía García-Alcalde, María Núñez-Sayar, Ander Ucelay-Aristi, Isabel Martínez Rodríguez.

Ethical Statements Human and animal rights statement and Informed consent The study was approved by the Local Clinical Research Ethics Committee (ethical approval code: 2023.013) and was carried out following the protocol and the standard procedures that ensure compliance with Declaration of Helsinki and Good Clinical Practice standards. Animals were not involved in this research. According to the Clinical Research Ethics Committee in our center and according to the Spanish local legislation, this type of retrospective research did not require informed consent in our country.

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